MIL-I-6533B
25 May 1970
SUPERSEDING
MIL-I-6533A
30 April 1951

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE MULTI-FUNCTION, TYPE G-10

This specification is mandatory for use by all departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of multi-function electrical resistance temperature indicator, designated Type G-10.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-I-7070	Indicator, Temperature, Electrical-Resistance Hermetically Sealed, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-831	Test Reports, Preparation Of
MS33558	Numeral And Letter Aircraft Instrument Dial, Standard Form Of
MS33639	Case, Instrument, Clamp-Mounted Aircraft
MS33585	Pointers, Dial Standard Design Of Aircraft Instrument
MS33678	Connector, Receptacle, Electric Integral Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6685

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* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

3. REQUIREMENTS

* 3.1 General specification. The requirements of MIL-I-7070 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

* 3.2 Design and construction.

* 3.2.1 Case. The indicator case shall conform to MS33639. The case length shall not exceed 4 inches.

3.2.2 Dial. The dial shall conform to Figure 1. The diameter of the dial as measured across the outer ends of the graduations shall be not less than 1-3/4 inches. The "0" degree indication shall be positioned on the horizontal centerline at the left side of the dial.

3.2.2.1 Scale arc. The indicator shall provide a minimum scale arc angle of 250 degrees. The mechanism shall not be geared to obtain the specified scale arc.

* 3.2.3 Pointer. The dimensions of the pointer tip shall be in accordance with MS33585.

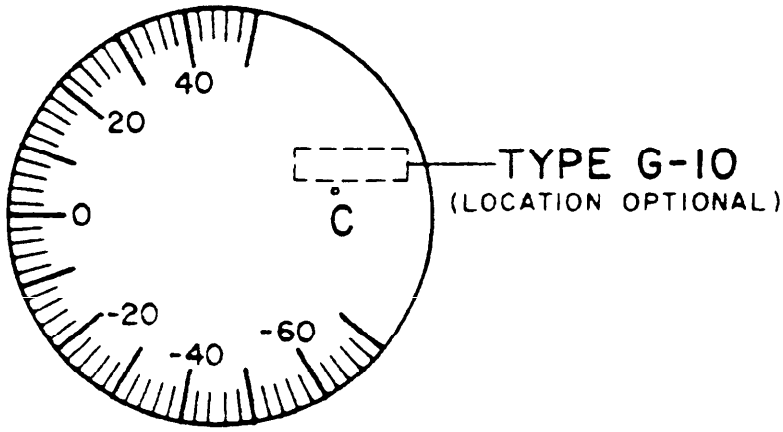
3.2.3.1 Pointer overswing. The pointer overswing shall be not less than 3 angular degrees beyond each end of the scale.

3.3 Markings.

3.3.1 Fluorescent-luminescent material. The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005
Numerals	0.125	0.020
Major Graduations	0.156	0.020
Minor Graduations	0.094	0.010
Lettering "°C"	0.125	0.020
Visible portion of pointer	- - -	- - -

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DIAL

FIGURE 1.

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3.3.2 Durable dull black. The marking "TYPE G-10," the background of the dial, and all other markings not specified shall be finished in durable dull black. The dimensions of the markings shall be 0.062 ±0.005 inch high, and 0.010 ±0.005 inch wide.

3.3.3 Mask. Durable dull black paint applied to the inner surface of the cover glass shall be considered acceptable as a mask. If the moving parts of the mechanism are effectively masked by the dial, no additional mask shall be required.

3.4 Weight. The weight of the complete indicator shall not exceed 1 pound.

* 3.5 Electrical receptacle. An MS33678-14S-2P electrical receptacle shall be provided on the rear of the indicator case. The receptacle shall be located on the vertical and horizontal centerline. The polarizing key of the receptacle shall be located at the top approximately on the vertical centerline. The receptacle shall be wired as follows:

Terminal	Connection
A	±24 Volt Battery
B	Resistance Bulb
C	Common - Negative 24 volt battery and remaining resistance bulb connection
D	Open

3.6 Identification of product.

3.6.1 Nameplate. The nameplate shall be marked with the following information:

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE,
MULTI-FUNCTION
Type G-10
Specification MIL-I-6533B
Manufacturer's Part No.
Manufacturer's Seral No.
Stock No. (USAF or Navy, as applicable)
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

* 3.7 Instrument identification decalcomanias. The indicator shall be provided with one each of the three identification decalcomanias shown on Figure 2. The decalcomanias shall be larger in diameter than the mask, but not of a size to interfere with readability of the indicator. The letters shall be finished in fluorescent-luminescent material. The background of the decalcomania shall be transparent or dull black. The letters shall be 0.125 ±0.010 inch high, and the width of the lines shall be 0.020 ±0.005 inch. The form of the letters shall conform to MS33558. The decalcomanias shall be the water-applied type and shall be suitable for application to the cover glass of the indicator. The decalcomanias, when mounted on the cover glass of the indicator, shall be capable of meeting the tests specified in Section 4 without peeling, blistering, fading, loss of adhesion, checking, softening, or hardening.

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IDENTIFICATION DECALCOMANIAS

FIGURE 2.

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3.7.1 Each set of decalcomanias shall be packaged in an envelope marked "THIS ENVELOPE CONTAINS IDENTIFICATION DECALCOMANIAS." Complete and detail instructions for mounting the decalcomanias on the cover glass of the indicator shall be furnished with each set of decalcomanias.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 General specification. Qualification and acceptance tests of MIL-I-7070 apply as requirements of this specification

* 4.1.2 Test report. When tests are conducted at a location other than the laboratory of the qualifying activity (procuring activity) a test report prepared in accordance with MIL-STD-831 shall be furnished to that activity.

4.2 Test conditions.

4.2.1 Test voltage. The test voltage shall be 28.5 ±0.5 volts direct current (DC), unless otherwise specified.

4.3 Test tolerances. The tolerances specified in Table I, not inclosed in parentheses, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, inclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.4 Inspection tests.

4.4.1 Individual tests.

4.4.1.1 Scale error at room temperature. The test resistance specified in Table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied, and the scale errors at the test points specified in Table I shall not exceed the tolerance specified in Table I.

4.4.1.2 Position error. The position errors of the indicator shall not exceed 1.5° Centigrade (C).

4.4.1.3 Friction error. The friction errors of the indicator shall not exceed 1.5°C.

4.4.1.4 Sealing. No leaks shall result from the sealing test.

4.4.1.5 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

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TABLE I

TEST TOLERANCES

Test Point	Test Resistance	Scale Error At Room Temperature	Reduced Voltage	Scale Error At Low Temperature	Scale Error At High Temperature	Scale Error At Room Temperature After Vibration	Scale Error At Room Temperature After Open And Short Circuit
°C	Ohms	Tol. ±°C	Tol. ±°C	Tol. ±°C	Tol. ±°C	Tol. ±°C	Tol. ±°C
-70	68.27	2.0 (2.0)	2.5 (2.5)	3.5 (3.5)	3.5 (3.5)	2.5 (2.5)	3.5 (3.0)
-50	74.24	1.5 (1.5)	0.5 (0.5)	3.0 (3.0)	3.0 (3.0)	2.0 (2.0)	3.0 (2.5)
-30	80.56	1.0 (1.0)	0.5 (0.5)	2.0 (2.0)	2.0 (2.0)	1.5 (1.5)	2.5 (2.0)
-10	87.04	1.0 (1.0)	0.5 (0.5)	2.0 (2.0)	2.0 (2.0)	1.5 (1.5)	2.5 (2.0)
0	90.38	1.0 (1.0)	0.5 (0.5)	2.0 (2.0)	2.0 (2.0)	1.5 (1.5)	2.5 (2.0)
+10	93.80	1.0 (1.0)	1.0 (1.0)	2.0 (2.0)	2.0 (2.0)	1.5 (1.5)	2.5 (2.0)
+30	100.91	1.5 (1.5)	1.0 (1.0)	3.0 (3.0)	3.0 (3.0)	2.0 (2.0)	3.0 (2.5)
+50	108.39	2.0 (2.0)	2.5 (2.5)	3.5 (3.5)	3.5 (3.5)	2.5 (2.5)	3.5 (3.0)

Test Point	Test Resistance	Scale Error With Bulb Lead Resistance	Extreme Low Temperature Exposure		
			Scale Error At Low Temperature	Scale Error At Room Temperature After Low Temperature	Scale Error At Room Temperature After High Temperature
°C	Ohms	Tol. ±°C	Tol. ±°C	Tol. ±°C	Tol. ±°C
-70	68.27	1.5 (1.5)	5.0 (3.5)	3.5 (2.0)	3.5 (2.0)
-50	74.24	1.5 (1.5)	4.5 (3.0)	3.0 (1.5)	3.0 (1.5)
-30	80.56	1.5 (1.5)	3.5 (2.0)	2.5 (1.0)	2.5 (1.0)
-10	87.04	1.5 (1.5)	3.5 (2.0)	2.5 (1.0)	2.5 (1.0)
0	90.38	1.5 (1.5)	3.5 (2.0)	2.5 (1.0)	2.5 (1.0)
+10	93.80	1.5 (1.5)	3.5 (2.0)	2.5 (1.0)	2.5 (1.0)
+30	100.91	1.5 (1.5)	4.5 (3.0)	3.0 (1.5)	3.0 (1.5)
+50	108.39	1.5 (1.5)	5.0 (3.5)	3.5 (2.0)	3.5 (2.0)

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4.4.2 Sampling tests.

4.4.2.1 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5 volts at the test points specified in Table I. The indications at the reduced voltage shall not differ from the indications with 28.5 volts applied by more than the tolerances specified in Table I.

4.4.2.2 Scale error at low temperature. The scale errors shall not exceed the tolerances specified in Table I.

4.4.2.3 Scale error at high temperature. The scale errors shall not exceed the tolerances specified in Table I.

4.4.2.4 Magnetic effect. The magnetic compass shall not deflect more than 5 degrees.

4.4.2.5 Resistance bulb current. A total resistance of 100 ohm shall be applied across the resistance bulb terminals of the indicator, the test voltage applied, and the resistance bulb circuit current measured. The current shall not exceed 22 milliamperes.

4.4.2.6 Vibration error. The pointer oscillation shall not exceed 1°C. The pointer variation shall not exceed 1.5°C.

4.4.2.7 Vibration failure. The scale errors after vibration shall not exceed the tolerances specified in Table I.

4.4.2.8 Open and short circuit. The scale errors after open and short circuit shall not exceed the tolerances specified in Table I.

4.4.3 Special tests.

4.4.3.1 Bulb lead resistance effect. The indicator shall be connected as shown on Figure 3 and subjected to a scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance by more than the tolerances specified in Table I.

4.4.3.2 Response time. The response time shall not exceed 3 seconds.

4.4.3.3 Damping factor. The damping factor shall be not less than 1.6.

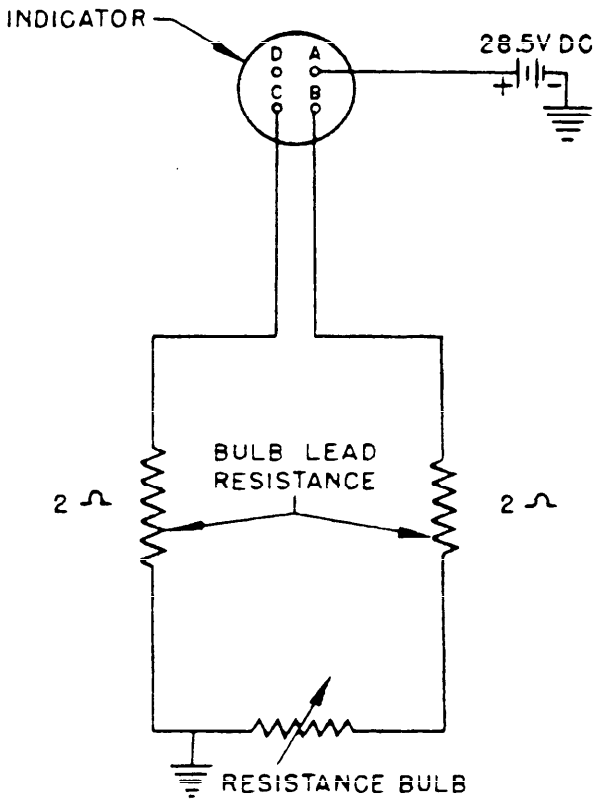
4.4.3.4 Power consumption. The power consumption shall not exceed 3 watts.

4.4.3.5 Extreme temperature exposure. The scale errors at low temperature shall not exceed the tolerances specified in Table I. The scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table I.

4.4.3.6 High temperature exposure. The scale errors at room temperature after high temperature exposure shall not exceed the tolerances specified in Table I.

4.4.3.7 Thermal shock. No damage to the indicator shall result from the thermal shock test.

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WIRING DIAGRAM

FIGURE 3.

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4.4.3.8 Salt spray. The indicator shall be subjected to and meet the requirements of the sealing test following salt spray exposure. The indicator shall not be subjected to a scale error test after salt spray. The insulation resistance shall be a minimum of 20 megohms.

* 4.5 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each indicator shall be preserved and packaged in accordance with MIL-P-116, Method III and placed in a snug fitting container conforming to PPP-B-636, Grade W6C.

* 5.1.2 Level C. Each indicator shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it meets this requirement.

* 5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping container conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall be limited to the applicable special requirements table of the specification. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the Appendix of PPP-B-636.

* 5.2.2 Level B. Same as level A except class domestic shipping containers may be used.

* 5.2.3 Level C. The indicators shall be packed in a shipping container which will be acceptable to the carrier at lowest rates and insure safe transportation to the point of delivery. Containers shall comply with Uniform Freight Classification Rules and regulations or other carriers as applicable to the mode of transportation.

* 5.3 Marking. In addition to any other markings required by order or contract, the interior and exterior markings shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type G-10 electrical resistance temperature indicator covered by this specification is intended to indicate temperatures in the range of -70° to +50°C, such as free air, carburetor air, and cabin air.

* 6.1.1 The type G-10 temperature indicator can be used on instrument panels requiring AN case mounting by the application of an instrument mounting bezel in accordance with MIL-B-6836.

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* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation, packaging and packing required (see Section 5).

* 6.3 General. The instructions contained in Section 6 of MIL-I-7070, including provisions for qualification tests, are applicable to and form a part of this specification.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Review Activities:
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Preparing Activity:
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Project No. 6685-0171

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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